

Ratul Ray

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SUMMARY

Postgraduate student pursuing a Master of Computer Applications (MCA) with specialization in Artificial Intelligence and Deep Learning (IBM Collaboration) at Assam Downtown University. Possesses a strong academic foundation in programming, machine learning, computer networking, and system logic. Demonstrated ability to design and implement AI-driven solutions through research projects and internships. Motivated to apply advanced AI and deep learning techniques to real-world problem statements.

SKILLS AND TOOLS

Java	Computer Networking	Android Studio
Python	REST APIs	Data Visualization
C Programming	MongoDB	GitHub

PROFESSIONAL EXPERIENCE

- International Business Machines Corporation (IBM)** July 2026 - August 2026
Summer Internship on FullStack Web Development
Gained hands-on experience in full-stack web development, working across frontend and backend technologies to design and develop responsive web applications. Strengthened skills in web application development, database integration, API implementation, debugging, and testing.
- Kareng Technologies**
Summer Internship on AIML and Python July 2023
Gained hands-on experience in Artificial Intelligence and Machine Learning using Python, with practical exposure to data preprocessing, model development, training, and evaluation. Strengthened skills in data analysis, machine learning techniques, and implementing AI-driven solutions to real-world problems.

PROJECTS

- JENNIE – AI Virtual Assistant**

Developing an AI-powered virtual assistant using React.js, TypeScript, JavaScript, HTML5, CSS3, Canvas API, Ollama, and Large Language Models (LLMs), featuring conversational and voice-based interaction with a responsive web interface. Implementing a multi-layer, state-reactive intelligence core with real-time visualizations for listening, processing, thinking, and speaking states, while continuously refining the system's natural-language understanding, contextual processing, and response generation capabilities.

- Real-Time Chat Application**

Designed and developed a real-time chat application supporting private and group messaging by following a structured development lifecycle from requirement analysis to deployment. The system features an interactive frontend built with HTML, CSS, and JavaScript, a Node.js and Socket.io–based backend for real-time messaging and user presence, and MongoDB for persistent storage of users, messages, and groups, ensuring low-latency communication and reliable chat history through WebSocket integration.

- A Review on the Current Trends in Crop Yield Forecasting Using Artificial Intelligence**

Examined modern methodologies and AI technologies utilized in predicting crop yields. Evaluated the efficacy of diverse machine learning models and data sources to offer recommendations for enhancing precision and efficacy in AI-driven crop yield forecasting.

[DOI- 10.36948/ijfmr.2024.v06i03.21432](https://doi.org/10.36948/ijfmr.2024.v06i03.21432)

- SURAKSHA AI**

Disaster management – risk mitigation, planning and management before, during or after a disaster.

Developed a real-time disaster risk intelligence platform using React, Vite, Leaflet Maps, Python, FastAPI, NDMA/SACHET data, river-gauge data, OpenStreetMap, and OSRM routing. Integrated 18 live APIs to continuously fetch and normalize government disaster warnings, map alerts to affected states and districts, calculate 0–100 risk scores, and provide interactive risk maps, safe-route recommendations, district-level insights, and situational reports without relying on a persistent database.

ACHIEVEMENTS

- **NASA Space Apps Challenge 2026** — Participated in the “Be An Earth System Trend Detective!” challenge, analyzing NASA Earth-system datasets to identify temporal and regional environmental trends, develop data-driven visualizations, and assess the statistical significance of observed changes.
- **Smart India Hackathon 2026** — Participated and developed SURAKSHA AI for SIH26206: Disaster Management, a real-time disaster risk intelligence platform designed for risk assessment, disaster monitoring, geospatial visualization, and response planning using live data sources and intelligent analytics.

EDUCATION

Master of Computer Applications (MCA)

Specialization: Artificial Intelligence and Deep Learning (in collaboration with IBM)

(2025-2027)

Assam Downtown University, Guwahati

Bachelor of Science in Information Technology

Major: Mobile Applications and Information Security

Grade: 8.04 CGPA

(2021 - 2024)

Assam Downtown University, Guwahati

Higher Secondary (Science)

Royal Public Senior Secondary School, Dhubri

HSLC

Shankardev Shishu Niketan, Gauripur

CERTIFICATIONS

Artificial Intelligence and Machine Learning with AIML

Kareng Technologies

Introduction to Generative AI by Google

Coursera

Android Development

Udemy

Python Developer

Udemy

Programming in C

Udemy
